

So,

$$\frac{\partial \mathcal{L}_{CE}}{\partial z} \propto (a-y) \quad \text{and} \quad \frac{\partial \mathcal{L}_{MSE}}{\partial z} \propto (a-y) \cdot a(1-a)$$

Notice that the cost for MSE has an extra factor of $a(1-a) = a - a^2$.

Taking $\frac{d}{da}(a - a^2) = 1 - 2a$, we find the max value of this function to be at $a = \frac{1}{2}$

$$(1 - 2a = 0 \rightarrow a = \frac{1}{2})$$

$$\text{So, } f\left(\frac{1}{2}\right) = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4}$$

Therefore $\forall a (a(1-a) \leq \frac{1}{4} \mid 0 < a < 1)$.

Thus to achieve a comparable rate of gradient descent (at best with $a \approx \frac{1}{2}$):

$$4 \cdot \eta_{CE} \gtrsim \eta_{MSE}.$$

Since values of a fluctuate around $0 < a < 1$, this factor of 4 is a best-case scenario, hence why our optimal learning rate for this model is 10x-100x larger than it would be using cross-entropy loss.